

Executable Disease Networks using CaSQ

Anna Niarakis and Sylvain Soliman

Inria

Computational Systems Biology for Complex Human Disease:
From static to dynamic representations of disease mechanisms



Sylvain Soliman

Computer Science Researcher

Inria Saclay, Lifeware team



Biography

Since 2002 I am a permanent researcher (CR) in the **Lifeware** (formerly Contraintes) group of Inria Saclay (formerly Paris-Rocquencourt).

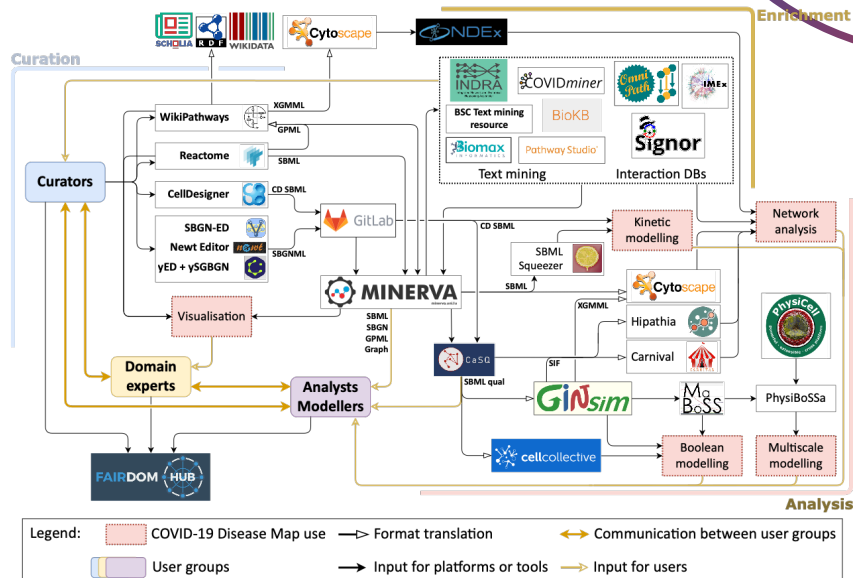
My research interests focus around Computational Biology and Theoretical Computer Science. In this context I'm one of the main developers and maintainers of the **BIOCHAM** platform. This is where most of the techniques I develop, using Constraint Programming, Model-Checking, and other formal methods get implemented.

On top of my BIOCHAM developments, you can find **below** other side projects concerning Vim, Prolog, etc. I used to maintain separately **Nicotine**, a constraint-based software for Petri-net invariant computation and **Tropical equilibration**, but it has now been completely merged into BIOCHAM.

I was teaching for about 15 years in the MPRI and for two years in the AI&AVC Master. You can find **below** more information.

I am president of Inria Saclay's **Scientific Commission**, and since 2016 I was on various Ph.D. grants committees. When I was in Rocquencourt I was president of the Doctoral Committee for quite some time, and of the Technological Development Commission for a few years.

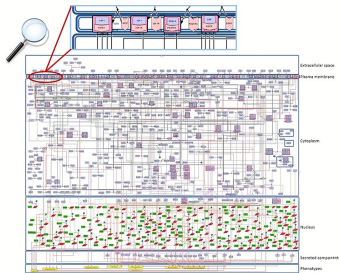
C19DM Ecosystem



Using CaSQ to bridge a gap...

People curate data/literature to build large, rich and complex **knowledge** maps

How do you **compare** them with **experiments**?

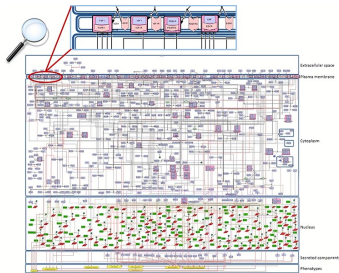


Map

Using CaSQ to bridge a gap...

People curate data/literature to build large, rich and complex **knowledge** maps

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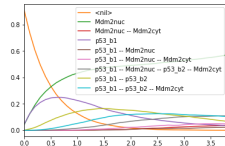
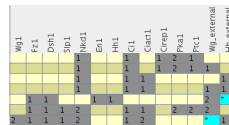
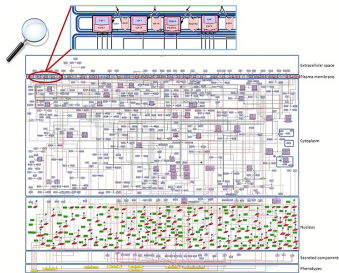
Map



Using CaSQ to bridge a gap...

People curate data/literature to build large, rich and complex **knowledge** maps

How do you **compare** them with **experiments**?



Map



Model

“Why build models?”

AFAIK Jay Bailey produces “plastic models”

<https://atlantis-models.com/why-build-models/>

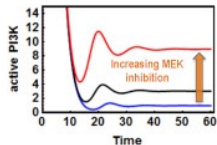
“To [...] **calculate logically** about what components and interactions are important in a complex system.”

What people call “digital twin” is often just a fancy name for “model”

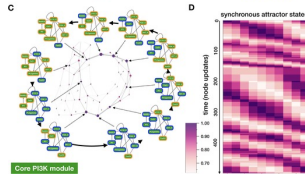
You already did yesterday!

Some tools:

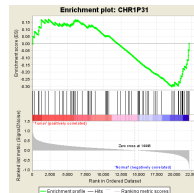
MaBoSS
CellCollective
BioModelAnalyzer
CoLoMoTo



Biological mechanisms
Explainability
**connecting
science**



Dynamic model



Pathways + enriched sets of genes

CellDesigner
Cytoscape



CaSQ

Static disease map

(Bottom-up) Molecular Interaction Maps

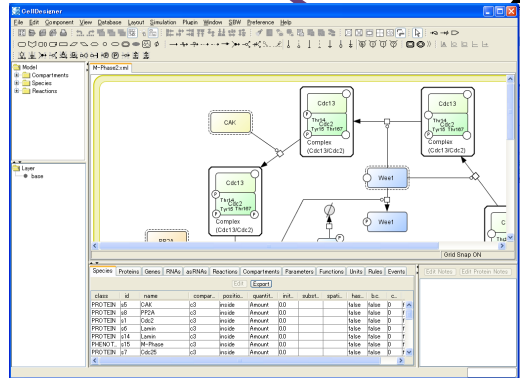
Built from scratch

Starting from an available map
(e.g., Rheumatoid Arthritis,
Covid-19 DM, Atlas of Cancer
Signalling Network)

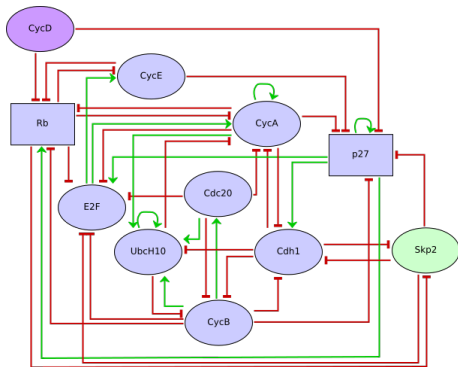
Reactome, KEGG → MINERVA
Translated into SBGN-PD

Described as *reusable* and *comprehensive* **Process Description**
diagrams: *detailed*, *mechanistic*

Encoded in SBML using CellDesigner: **rich annotations**



Logical Models *à la* René Thomas



Qualitative abstraction:

- Discrete (*Boolean*) state
- **Dynamics** $\Leftarrow$ logical formulae
- *Logical* time
- Phenotype $\equiv$ *steady state*

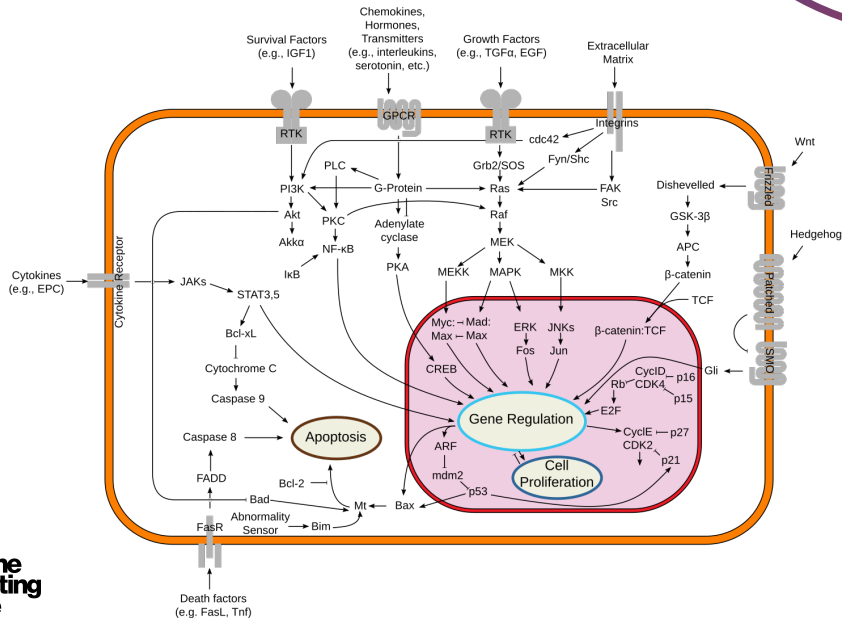
What CaSQ does

- Handle even very large detailed mechanistic maps
- Keep and use all information in the annotations
- Build a fully executable/dynamic logical model

What CaSQ does not do

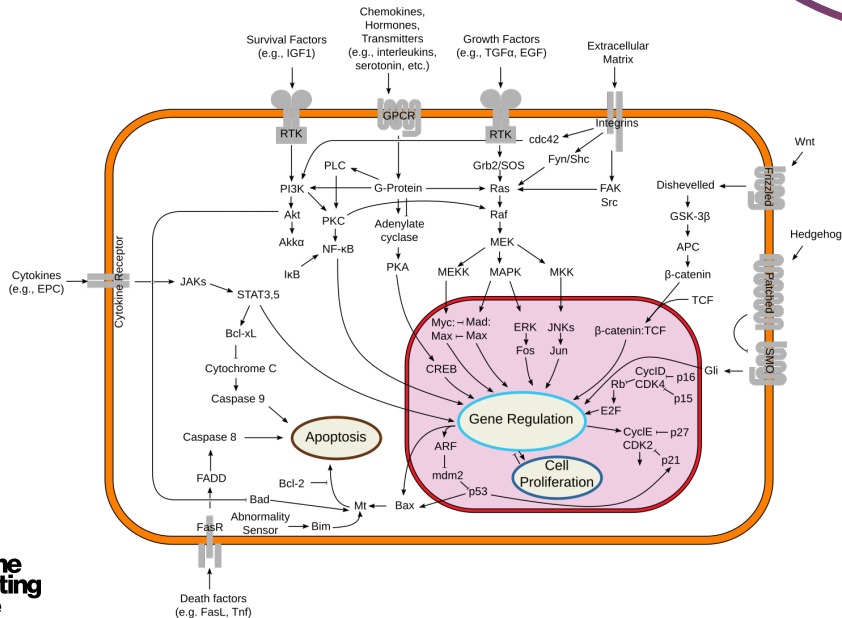
- extract a *relevant* submap
(CaSQ can extract up/down-stream however)
- only convert PD to AF
(he who can do the most can do the least,
CaSQ saves the AF as .sif file for Cytoscape/Hipathia)
- build a numerical (ODE-based) dynamic model
- build *the best* logical model
- handle the analysis of the model

Is this a “good” map?

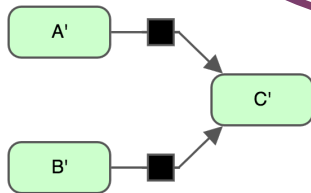
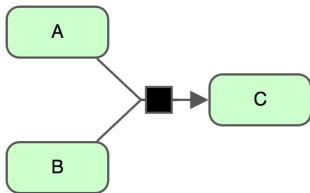


Is this a “good” map?

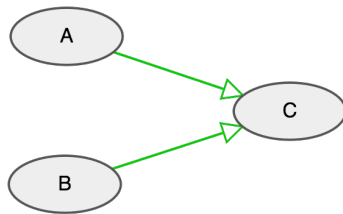
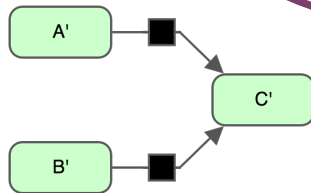
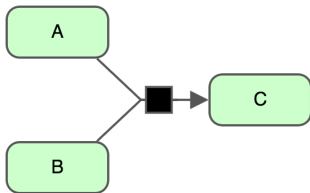
PD or AF?



The AF graph loses information

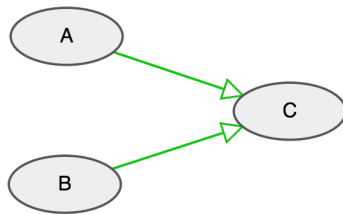
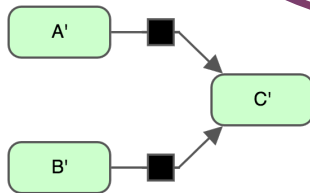
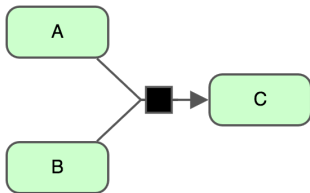


The AF graph loses information



Same Activity Flow,

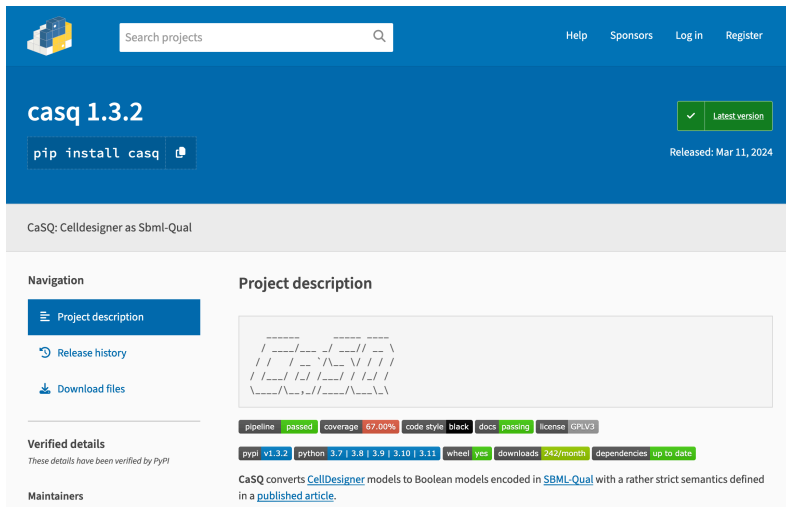
The AF graph loses information



Same Activity Flow, but different Logical Models ($\wedge$ vs. $\vee$)

Installing CaSQ (from PyPI)

```
python3 -m pip install casq
```



The screenshot shows the PyPI project page for **casq 1.3.2**. The page has a blue header with the project name and version, a search bar, and navigation links (Help, Sponsors, Log in, Register). Below the header, there's a green button with a checkmark and the text "Latest version". A button labeled "pip install casq" is also visible. The release date "Released: Mar 11, 2024" is shown. The project description "CaSQ: CellDesigner as SbmL-Qual" is displayed. On the left, a "Navigation" sidebar lists "Project description", "Release history", and "Download files". The "Project description" section features a diagram of a cell model and a table of project statistics. The statistics table includes: pipeline (passed), coverage (67.00%), code style (black), docs (passing), license (GPLV3), pypi (v1.3.2), python (3.7 | 3.8 | 3.9 | 3.10 | 3.11), wheel (yes), downloads (242/month), and dependencies (up to date). The project description text states: "CaSQ converts [CellDesigner](#) models to Boolean models encoded in [SBML-Qual](#) with a rather strict semantics defined in a [published article](#)."

casq 1.3.2

pip install casq

Released: Mar 11, 2024

CaSQ: CellDesigner as SbmL-Qual

Navigation

- Project description
- Release history
- Download files

Verified details

These details have been verified by PyPI

Maintainers

Project description

CaSQ converts [CellDesigner](#) models to Boolean models encoded in [SBML-Qual](#) with a rather strict semantics defined in a [published article](#).

pipeline	passed	coverage	67.00%	code style	black	docs	passing	license	GPLV3
pypi	v1.3.2	python	3.7 3.8 3.9 3.10 3.11	wheel	yes	downloads	242/month	dependencies	up to date

Usage (CLI or directly from Python, e.g., in CoLoMoTo)

```
uvx casq -h
```

```
usage: casq [-h] [-v] [-D] [-c] [-s] [-r S] [-f FIXED] [-n]
            [-u [UPSTREAM ...]] [-d [DOWNSTREAM ...]] [-b] [-g GRANULARITY]
            [-i INPUT] [-C]
            [infile] [outfile]
```

Convert CellDesigner models to SBML-qual with a rather strict semantics.
Copyright (C) 2019, Sylvain.Soliman@inria.fr GPLv3

positional arguments:

infile	CellDesigner File
outfile	SBML-Qual/BMA json File

options:

-h, --help	show this help message and exit
-v, --version	show program's version number and exit
-D, --debug	Display a lot of debug information
-c, --csv	Store the species information in a separate CSV (and .bnet) file
-s, --sif	Store the influence information in a separate SIF file
-r, --remove S	Delete connected components in the resulting model if their size is smaller than S. A negative S leads to keep only the biggest(s) connected component(s)
-f, --fixed FIXED	A CSV file containing input values or knock-ins/knock-outs, one per line, with name in the first column and

A 3-step process

- map simplification (4 rules)
- logical model (logical rules & graph) computation
- model simplification

[S. S. Aghamiri et al. Bioinformatics 36 (16) Aug. 2020]

Step 2 – General Logical Rule

Using *inhibitor* and other modifier annotations:

"A target is on when **any** of the reactions producing it is on
a reaction is on if **all reactants are on**, **all inhibitors are off** and
one of the catalysts is on if there are any."

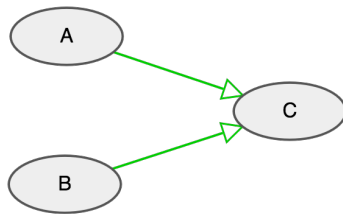
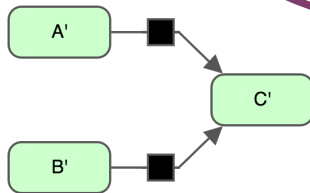
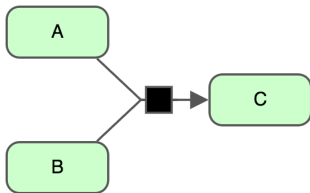
$$t = \bigvee_{R,C,I \rightarrow t \dots} \left(\bigwedge_{r \in R} r \bigwedge_{i \in I} \neg i \bigvee_{c \in C \text{ or } \{T\} \text{ if } C = \emptyset} c \right)$$

R POSITIVE t

C POSITIVE t

I NEGATIVE t

The AF graph loses information



Same Activity Flow, but different Logical Models ($\wedge$ vs. $\vee$)

Step 3 – Cleanup

Maps are **not connected**

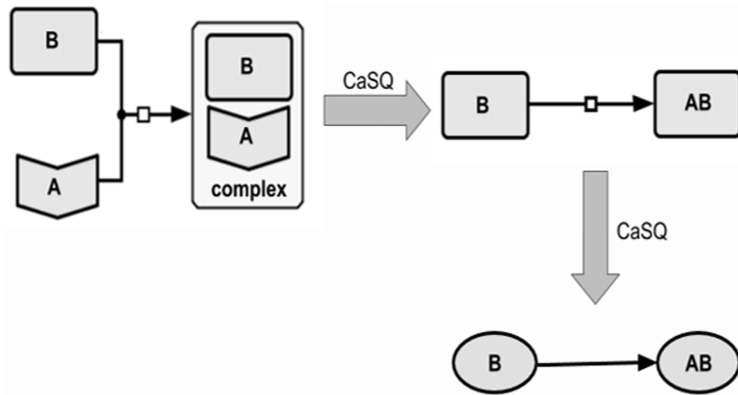
Keep only some CCs (according to their size)

Names are **ambiguous**

Rename species if necessary (e.g., type, modifications and compartment disappear in the model)

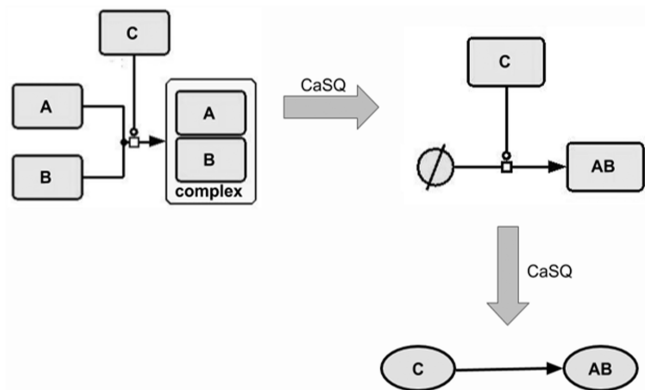
Generate auxiliary files if requested (SIF, CSV, BMA, BNET...)

Step 1, Rule 1 – Receptor Deletion



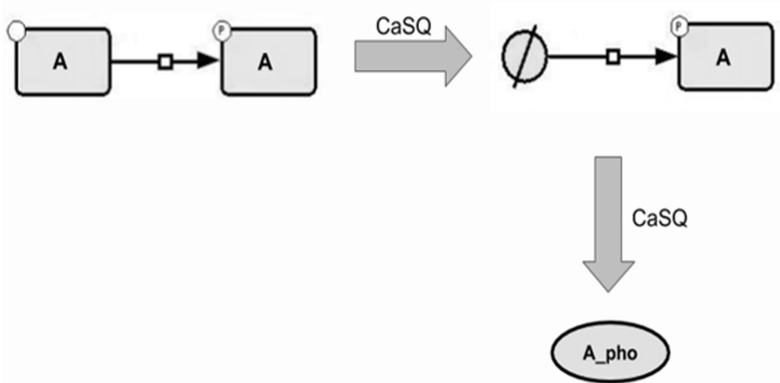
A and B do not take part in any other reaction. A annotated as **receptor**, reaction as *heterodimer association*

Step 1, Rule 2 – Complex Merge



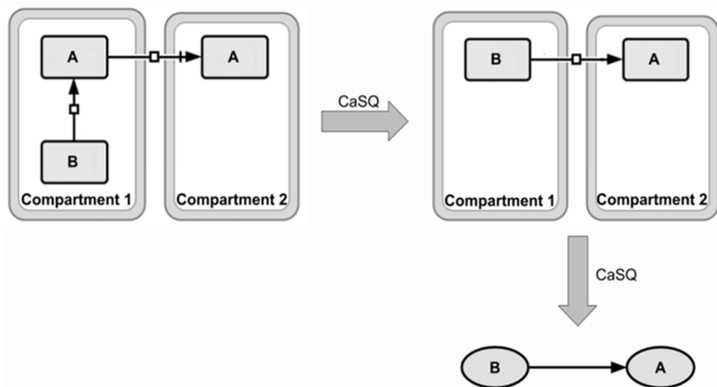
A and B do not take any part in any other reaction, reaction annotated as **heterodimer association**

Step 1, Rule 3 – Inactive Species Removal



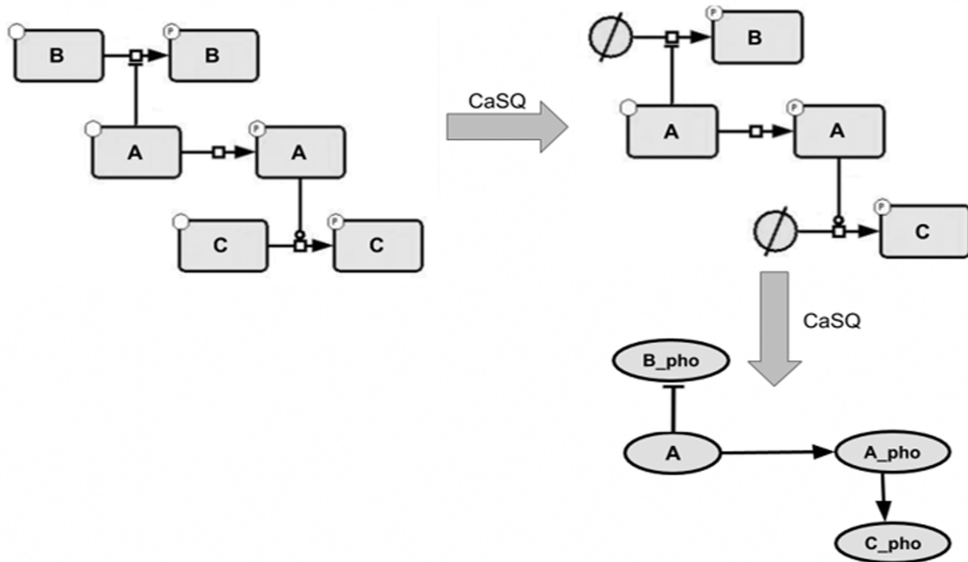
A only appears in this reaction with a single product, both reactant and product have the **same name**

Step 1, Rule 4 – Transport Simplification

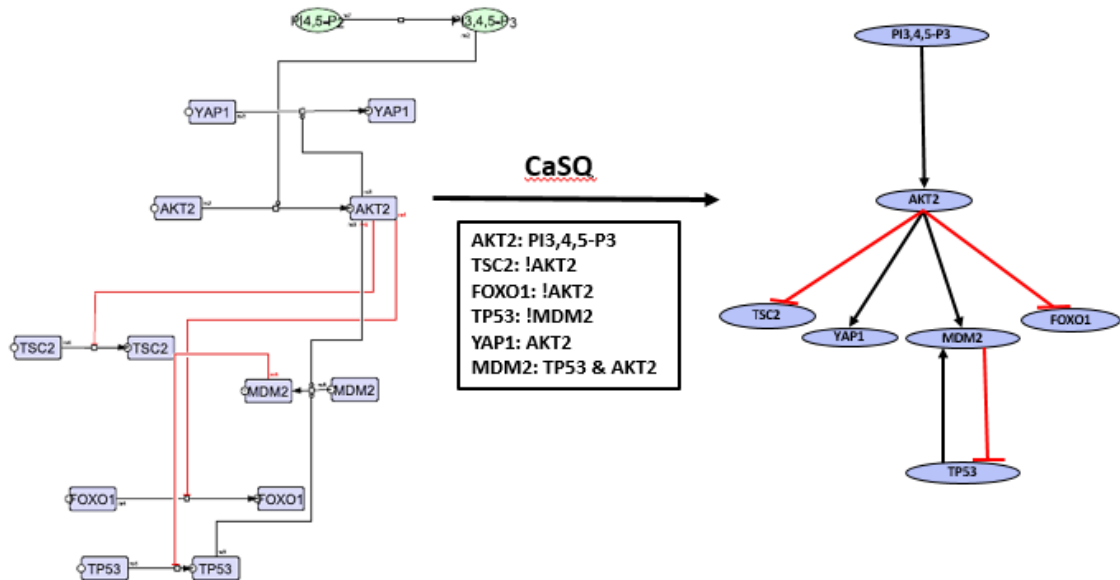


A (reactant) does not take any other *active* part, reaction annotated as **transport**, *same name* in both compartments

Example (practical reduction is about 30%)



RA map excerpt (50% reduction)



Conserving *information*

All vertices of the model correspond to one (species) vertex of the original map

Let us **keep its original layout** for readability!

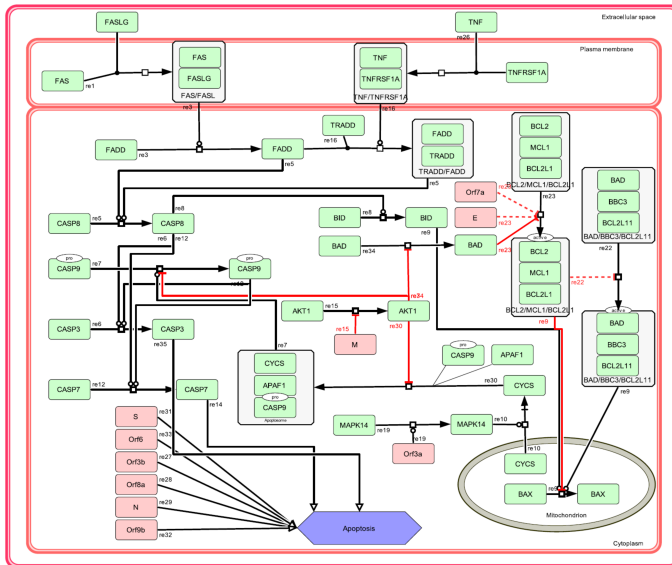
Annotations about the nature of a species (e.g., receptor) are used in the transformation

But there is much more, e.g., **bibliographic information**

MIRIAM: Minimal Information Required In the Annotation of Models

Propagate that of vertices that are kept. **Merge** that of removed vertices into a nearby meaningful vertex

Apoptosis module (C19DM)



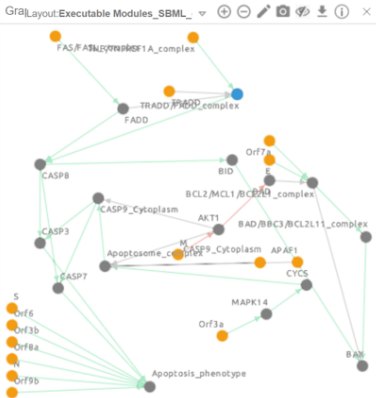
Apoptosis model in TCC

Executable Modules_SBML_qual_build_sbml_Apoptosis_stable.sbml (-3) - changes not saved

File Insert Edit Workspace Help

1.0 Overview Model Simulation Analysis Network Analysis Knowledge Base

GraphLayout:Executable Modules_SBML_



Internal Components

Name	0	1
AKT1	0	1
Apoptosis_phenotype	8	0
Apoptosome_complex	1	0
BAD	0	1
BAD/BBC3/BCL2L1_complex	1	0
BAX	2	0
BCL2/MCL1/BCL2L1_complex	2	0
BID	1	0
CASP3	2	0
CASP7	2	0

External Components

Name
APAF1
CASP9_Cytoplasm
E

Regulatory Mechanism TRADD/FADD_complex

Positive Regulators: Drop Component

Negative Regulators: Drop Component

Conditions

Condition

If/When FADD and TRADD are Active

(Co-operative)

SubConditions

Knowledge Base TRADD/FADD_complex

Description

Regulatory Mechanism Summary

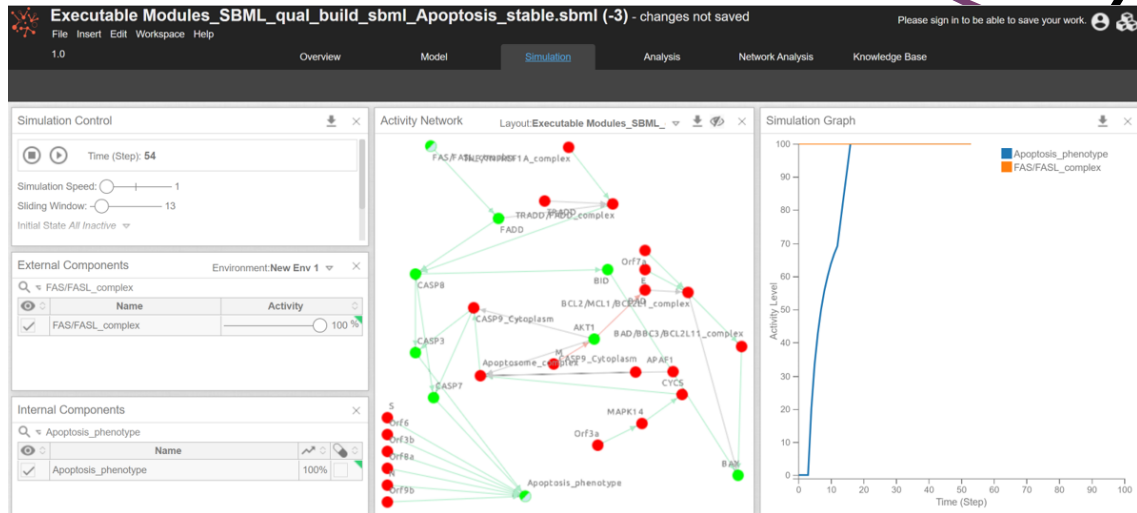
Upstream Regulators

FADD

TNF/TNFRSF1A_complex

TRADD

Apoptosis model in TCC



SBML/SBGN in, SBML-qual out

Standards are crucial for interoperability

Rich annotations “in”



Rich annotations “out”



Conclusion

CaSQ automatically produces a
dynamic executable model from a static map

It strongly relies on annotations & topology to infer logical rules

It embraces SBML standards in and out for interoperability

It conserves rich annotations and layout, even when reducing the model, for readability and reusability

This does not lead to a **perfect** model, but is scalable, fast and consistent for a first step

"The map is not the territory."
— Alfred Korzybski

"The map is not the territory."
— Alfred Korzybski

"All models are wrong, but some
are useful."
— George Box

C19DM Ecosystem

